

Downstairs can be bright—but fixture output matters.

Calculation conclusion: build in a useful full-output setting, then dim for evening. The current unselected fixture schedule does not establish that capacity.

RESULT / A CLEAR DESIGN DIRECTION

FAMILY ROOM / 356 SF	DIRECT AVERAGE	WITH FINISH SENSITIVITY
6 broad fixtures / 1000 class	91 lux	98–109 lux
6 broad fixtures / 1500 class	118 lux	127–142 lux
8 broad fixtures / 1500 class	166 lux	178–196 lux

The eight-light option comes close to a 200-lux bright general-light setting without any daylight or sconces. Six stronger broad fixtures can support a softer room, but need a quantified supplemental layer to reach the same brightness.

My recommendation

Use the eight-point broad-beam layout as the brightness benchmark. If a quieter ceiling is more important, retain six and have a real sconce/wall-light package calculated to make up the shortfall. Coordinate all ceiling openings with the acoustic assembly.

What was actually calculated

Manufacturer IES intensity data, an 8-foot ceiling, and point-by-point illumination. The family room uses the full A1-01 356-square-foot area, not the smaller room-label box in the tour.

What “1500 class” means

It is the manufacturer’s nominal package name. The chosen 2700K white-trim IES file integrates to about 1,094 delivered lumens. The calculation uses that actual intensity data, then a 0.80 maintenance factor.

What this is not

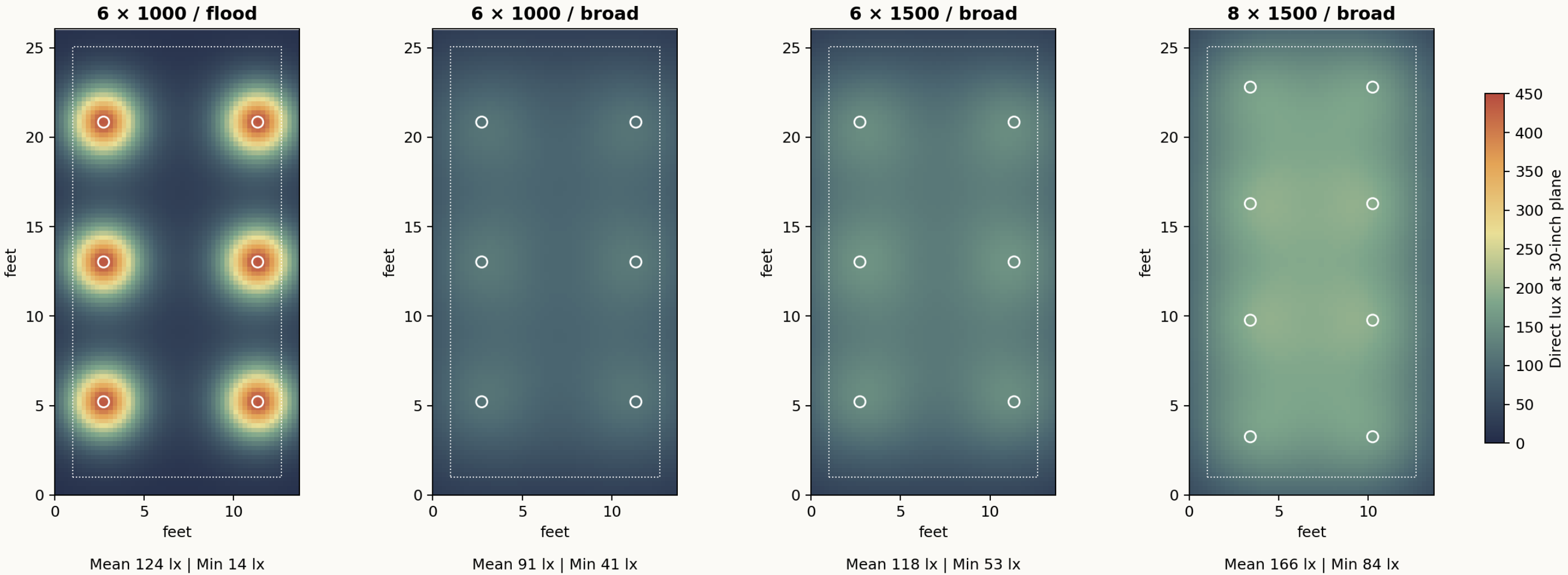
No fixture has been ordered or finally selected. The DMF 4-inch basis is a performance reference; any preferred 2–3-inch substitute needs its own matching photometry.

Sources: A1-01 / A1-03 / A3-04, Aug. 28 set; DMF M Series 2700K GA white-trim IES files. Values are maintained electric-light estimates, not measurements.

Codex | September 25, 2026 | Design study with candidate photometry; final fixtures, placement and construction coordination remain open.

Coverage is as important as the average.

Family-room comparison: the same full room, the same 8-foot ceiling, and a 30-inch evaluation plane. No reflected light is credited in these maps.

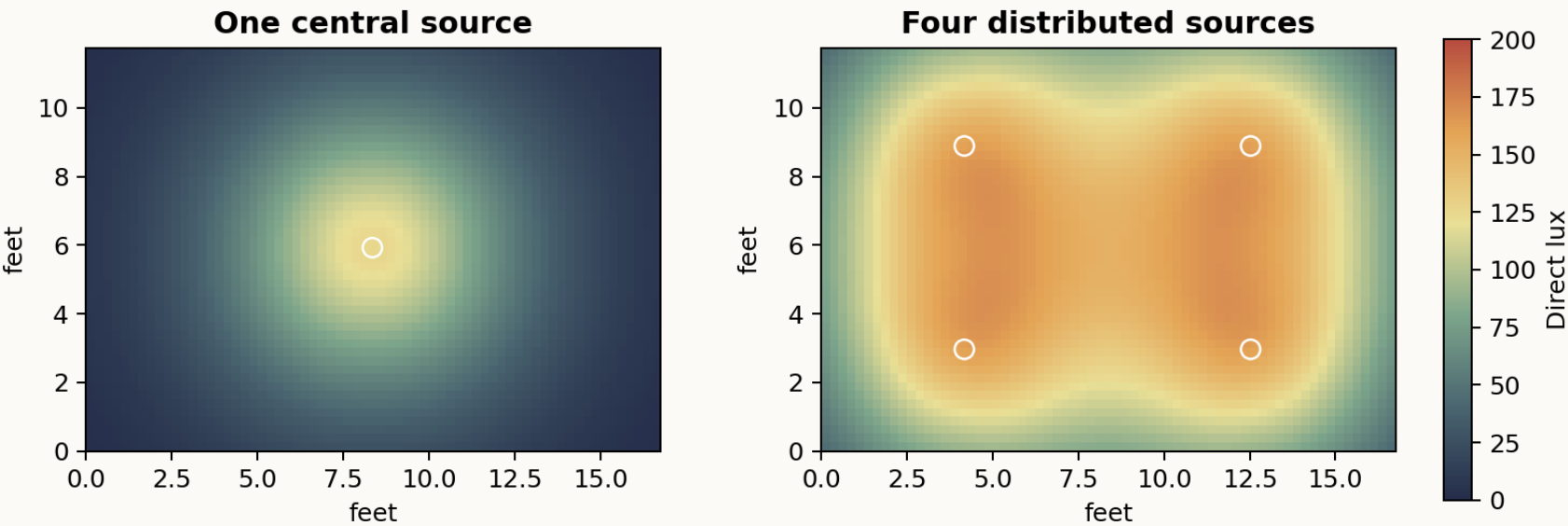


WHAT THE MAPS SHOW	DESIGN CONSEQUENCE
Narrower flood: 14–440 lux in the evaluated zone	Bright pools can coexist with dim gaps. Do not buy on lumen count alone.
Eight broad sources: 84–198 lux before reflections	More even general light; reading, work and the wet-bar counter still need task lighting.

White rings = candidate fixture centres; dotted line = statistical boundary 1 ft inside walls. Common color scale 0–450 lux; 1 foot-candle ≈10.764 lux.

Bedrooms: give the room ambient light and the task its own light.

A central downlight is a poor stand-in for a complete bedroom scheme. Actual fan lights and decorative fixtures remain unselected.



ROOM	TEST BASIS	EST. AVG*
Bedroom 1	4 × 1500 broad	163–181 lx
Bedroom 2	4 × 1000 broad	142–159 lx
Bedroom 3 / WFH	4 × 1500 broad	150–166 lx
Bedroom 4 / office	4 × 1000 broad	155–176 lx
Gym	4 × 1500 broad	198–224 lx

Bedroom 3 example: one central 1500-class downlight averages only 49 direct lux. Four distributed 1500-class sources give 141 direct lux and about 150–166 lux with the tested finishes. A diffuse ceiling fixture could be another solution; its actual IES data must be checked.

*Finish sensitivity: dark to light room assumptions. These are proposed capacity tests, not issued fixture counts. Sources are dimmable; high-output capacity is not the intended bedtime setting.

Aim for 300–500 lux on the activity itself

These are my study targets for reading, desk work and detailed tasks—not a requirement to illuminate the entire bedroom that brightly. Use selected bedside/desk lights and verify the actual working surface.

Areas: A1-01—Bed1 179, Bed2 149, Bed3 198, Bed4 130, Gym 135 sf. Rectangular test proportions are approximate; exact furniture and ceiling placement need coordination.

The remaining first-floor zones need different solutions.

Light the circulation floor, the laundry worktop and faces at mirrors separately. One room-wide brightness number cannot stand in for all three.

ZONE	CALCULATED BASIS	DIRECT / FINISH AVG	CONCLUSION
Hall 101 / breezeway	2 × 1000 broad; floor plane	71 / 82–106 lx	Useful baseline; light finishes help. Confirm final entry/door geometry.
Hall 103	3 × 1000 broad; floor plane	79 / 90–116 lx	Suitable circulation starting point around the 100-lux study target.
Hall 108	2 × 1000 broad; floor plane	73 / 84–112 lx	Similar circulation basis; closet light is a separate task.
Laundry	4 × 1500 broad; 36-inch plane	274 / 296–342 lx	Good general folding/work-surface basis; cabinet shadows may need a task strip.
Bath 1	2 × 1500 broad; 30-inch plane	142 / 154–179 lx	Ambient layer only. Mirror-face and shower/niche output still need actual fixtures.
Bath 2	2 × 1500 broad; 30-inch plane	170 / 184–219 lx	Ambient layer only; use separate mirror lighting and a wet-location fixture layout.
Closet 112	1 × 750 broad; floor plane	33 / 39–57 lx	Floor test is modest; illuminate hanging clothes/shelf fronts with integrated light.
Garage	6 × 1500 downlight surrogates	82 / 87–93 lx	Too little for the 200-lux study target. Select the six shop bars by delivered output.

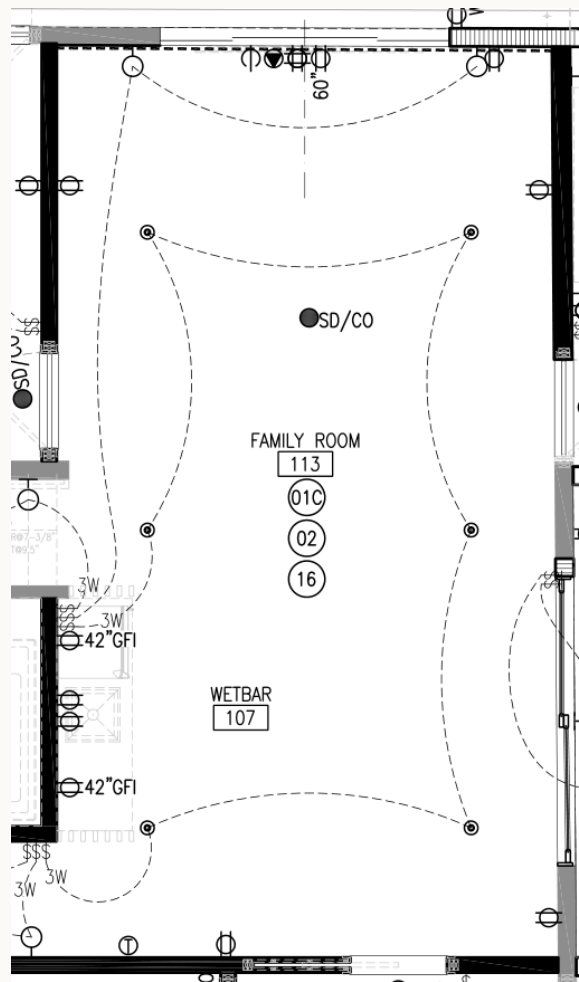
| Garage output budget + uncalculated zones

For 633 sf, six 4,500-lumen shop fixtures estimate about 165–239 average lux using a 0.45–0.65 utilization factor and 0.80 maintenance factor. That is a lumen-method budget, not an IES layout. Stairs, utility closet, mirror faces, shelf fronts and covered patio require their own selected-fixture checks.

Targets are Codex planning criteria, not code findings. Utility-room point layouts are proposals. Garage shop bars, decorative lights and vertical task surfaces are not selected.

A calculation you can hand to your architect and electrician.

The source files, coordinates, assumptions and numerical results are saved beside this report so the fixture supplier can reproduce or replace the model.



PLAN PATTERN / SIX F2 POINTS

The reviewed A1-03 shows six recessed points in the family/wet-bar area. The eight-point option redistributes the layout; it is not simply two extra holes at arbitrary locations. Confirm framing, fan/speaker positions and the resilient-channel detail.

Inputs and assumptions

8-foot ceiling; nominal 2700K; manufacturer Type-C candela data; 0.80 light-loss factor. Horizontal grids: 30 inches for living/bedrooms/baths, 36 inches for laundry, floor level for halls/closet. Quarter-foot direct grid; reported statistics exclude a one-foot perimeter.

Finishes and limits

Supplemental diffuse-box tests use ceiling/wall/floor reflectances of 80/50/20% and 80/20/15%. These are assumptions, not measured paint samples. No daylight, furniture, people, specular reflections or detailed window transmission is modeled. Shadows may lower local light; daylight adds a separate benefit.

Checks completed

On-axis output matches $\text{candela} \div \text{distance}^2$; doubling distance quarters illuminance. Diffuse exchange preserves area reciprocity and limits energy gain. Refining family-room surface patches from 1.5 to 1 foot changed mean light-case output by 0.7%. This checks numerical stability, not field accuracy.

Before final fixture release

Match the chosen housing, trim, color temperature, beam and dimmer to its IES file. Verify the exact room outline and centres; recalculate task planes and shadows; approve a physical mockup. Keep bright-day, evening and night-route layers independently controllable.

SOURCE LINKS: dmflighting.com/product/drd-m-series-4in-downlight/ · A1-01/A1-03/A3-04 in Aug. 28 revision set. Full URLs, IES files, CSVs and assumptions are in README.md and source_manifest.json.